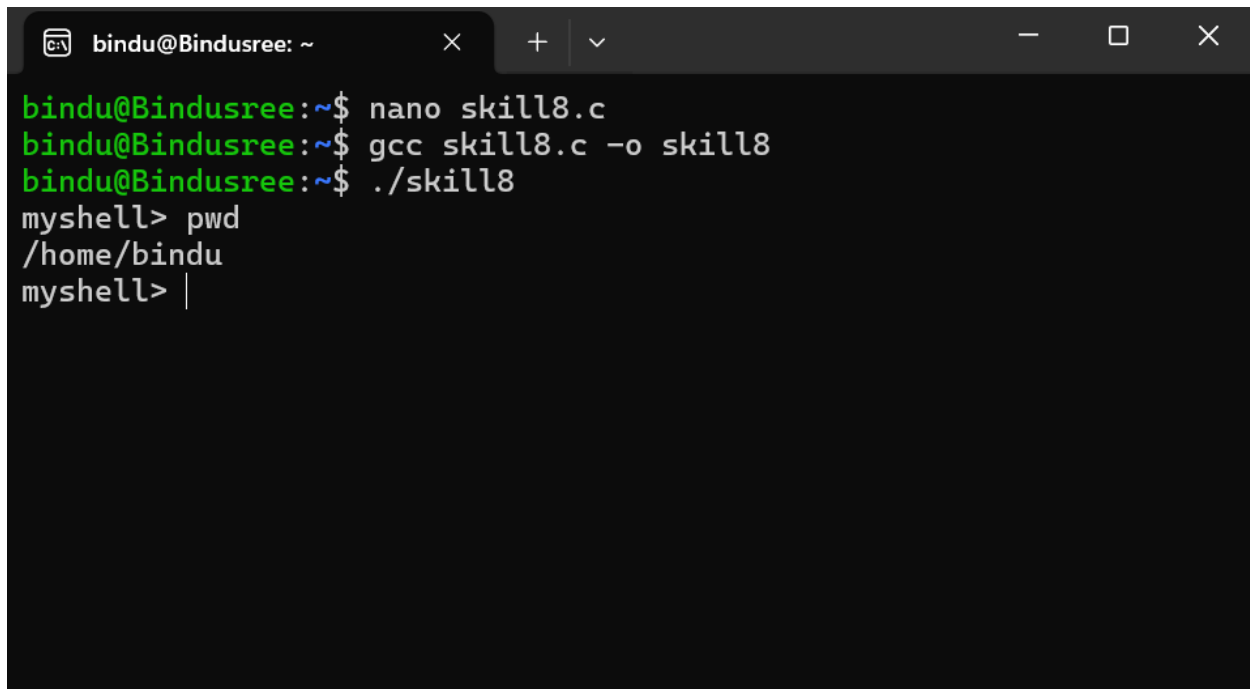


8.o Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

A terminal window with a dark background and light green text. The window title bar shows 'bindu@Bindusree: ~' and standard window controls. The terminal content shows a sequence of commands and their outputs: 'nano skill8.c' is entered; 'gcc skill8.c -o skill8' is entered; './skill8' is entered; 'myshell> pwd' is entered, followed by the output '/home/bindu'; and finally 'myshell> |' is entered.

```
bindu@Bindusree:~$ nano skill8.c
bindu@Bindusree:~$ gcc skill8.c -o skill8
bindu@Bindusree:~$ ./skill8
myshell> pwd
/home/bindu
myshell> |
```

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        printf("myshell> ");
        fgets(input, sizeof(input), stdin);

        input[strcspn(input, "\n")] = '\0';

        // Exit built-in command
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        // pwd built-in command
        if (strcmp(input, "pwd") == 0) {
            char *pwd = getenv("PWD");

            if (pwd != NULL)
                printf("%s\n", pwd);
            else
                printf("PWD variable not found.\n");

            continue;
        }

        // Variable expansion: echo $VARIABLE
        if (strncmp(input, "echo $", 6) == 0) {
            char *variable = input + 6;
            char *value = getenv(variable);

            if (value != NULL)

```

```
    char *variable = input + 0,  
    char *value = getenv(variable);  
  
    if (value != NULL)  
        printf("%s\n", value);  
    else  
        printf("Undefined variable: %s\n", variable)  
        continue;  
}  
  
    printf("Unknown command: %s\n", input);  
}  
  
return 0;  
}
```